

CS471 – Lesson 18 In-Class Worksheet

Probability: Uncertainty, Conditioning, and Independence

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Interpret probability as tactical uncertainty, compute joint and conditional probabilities, and evaluate statistical independence.

1. Joint Probability Distribution and Marginalization

An automated Early Warning Radar monitors an airspace corridor. The system tracks two random variables: Target Presence ($T \in \{\text{Target}, \text{NoTarget}\}$) and Radar Alert status ($A \in \{\text{Alert}, \text{NoAlert}\}$).

The full joint probability distribution $P(T, A)$ is given below:

	$A = \text{Alert}$	$A = \text{NoAlert}$
$T = \text{Target}$	0.04	0.01
$T = \text{NoTarget}$	0.05	0.90

- Verify that this table represents a valid probability distribution.
- Compute the marginal probability that an actual target is present, $P(T = \text{Target})$, by marginalizing (summing out) over variable A .
- Compute the marginal probability that the radar triggers an alert, $P(A = \text{Alert})$, by marginalizing over variable T .

2. Conditional Probability and Belief Updating

Using the radar joint distribution from Section 1:

- Write the mathematical formula for the conditional probability $P(T = \text{Target} \mid A = \text{Alert})$.

- b. If the radar sounds an alarm ($A = \text{Alert}$), compute the updated probability that an actual enemy target is present. Explain why this value is lower than the radar's raw detection rate.

3. Evaluating Statistical Independence

- a. State the mathematical definition of statistical independence between two events A and B .

- b. Using the joint probability table from Section 1, determine whether Target Presence (T) and Radar Alert (A) are statistically independent. Show your calculations.

Part 2: Student Independent Practice

Objective: Apply marginalization, conditional probability inference, and independence testing in defense operational domains.

A. Anti-Submarine Warfare (ASW) Acoustic Classification

An autonomous sonobuoy array analyzes acoustic contacts in a maritime chokepoint. The system tracks Contact Classification ($C \in \{\text{Hostile}, \text{Biologic}\}$) and Acoustic Frequency Profile ($F \in \{\text{LowFreq}, \text{HighFreq}\}$):

	$F = \text{LowFreq}$	$F = \text{HighFreq}$
$C = \text{Hostile}$	0.12	0.03
$C = \text{Biologic}$	0.18	0.67

1. Calculate the marginal probability of detecting a Low-Frequency contact, $P(F = \text{LowFreq})$.
2. Calculate the marginal probability that the contact is Hostile, $P(C = \text{Hostile})$.
3. If the sonobuoy detects a Low-Frequency acoustic profile ($F = \text{LowFreq}$), compute the updated conditional probability that the contact is Hostile, $P(C = \text{Hostile} \mid F = \text{LowFreq})$.

B. Electronic Warfare & Redundant Data Links

A command post operates two encrypted communications links: Link 16 (L_1) and SATCOM (L_2). The probability of Link 16 dropping due to adversary jamming is $P(L_1 = \text{Fail}) = 0.20$. The probability of SATCOM dropping due to severe weather is $P(L_2 = \text{Fail}) = 0.10$.

Assume the physical causes of failure for the two links are completely independent.

1. Compute the joint probability that both communication links fail simultaneously, $P(L_1 = \text{Fail}, L_2 = \text{Fail})$.

2. Compute the probability that at least one communication link remains operational.

C. Conceptual Analysis: Conditional Independence

An autonomous reconnaissance drone carries an Infrared (IR) camera and an Acoustic microphone. Both sensors detect an enemy generator starting.

- Explain why the two sensor alerts are **dependent** overall.
- Explain why the two sensor alerts become **conditionally independent** once the true operational state of the generator is known.